FINITENESS OF A MODULE DESCENDS ALONG A RING MAP INDUCING ISOMORPHISMS ON QUOTIENTS BY POWERS OF AN ELEMENT

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Allowed standard prerequisites (editorial).

- The Beauville-Laszlo faithful descent statement for modules killed by tensoring with R' oplus R_f. invoked at proof lines 25211-25212. Converts the finiteness of the cokernel condition into finite generation of M.
- The hypothesis that R -> R' induces isomorphisms R/f^nR -> R'/f^nR'. used at statement lines 25194-25195. Places R' with R_f as a cover so that the descent statement applies.
- The same-quotients statement identifying the completion case with the general one. invoked at proof lines 25213-25214. Derives the stated example about R^wedge from the main statement.

Source reference. Original lemma 25149: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25149–25149. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 25129: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25129–25129. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring, let $f \in R$, and let $R \to R'$ be a ring map which induces isomorphisms $R/f^nR \to R'/f^nR'$ for $n > 0$. An R-module M is finitely generated if and only if the $(R' \oplus R_f)$-module $M \otimes_R (R' \oplus R_f)$ is finitely generated.*

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For example, if $M \otimes_R (R^\wedge \oplus R_f)$ is finitely generated as a module over $R^\wedge \oplus R_f$, then M is a finitely generated R-module.

Proof. The 'only if' is clear, so we assume that $M \otimes_R (R' \oplus R_f)$ is finitely generated. In this case, by writing each generator as a sum of simple tensors, $M \otimes_R (R' \oplus R_f)$ admits a finite generating set consisting of elements of M. That is, there exists a morphism from a finite free R-module to M whose cokernel is killed by tensoring with $R' \oplus R_f$; we may thus deduce M is finite generated by applying Lemma 25149 to this cokernel. The last statement is a special case of the first statement by Lemma 25129. □